

POSTER: GEM-PLI SIDE CHANNEL IN GPON INFERRING ENCRYPTED STREAMING TRAFFIC

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INTRODUCTION & MOTIVATION

- 1.As fiber internet access has become dominant in Network Infrastructures passive optical network (PON) technologies-accounted for 54.2% of Fiber To The Home (FTTH) market in 2024 it is important to ensure traffic confidentiality.
- 2.GPON supports AES-based downstream traffic encryption on GEM Layer.
- 3.However, some fields of GEM Header remain plaintext
- 4.recent studies have demonstrated that such metadata like PLI can be used to infer victim's activity.
- 5.we researched whether the plaintext PLI field in GPON GEM header can be used to leak victim's privacy on the same optical splitter

BACKGROUND

GPON Topology and GTC Framing GPON Networks consist of an Optical Line Terminal (OLT) at the ISP and multiple Optical Network Terminals/Units(ONTs/ONUs) (typically called "modem") deployed at subscribers. A single optical tree can serve up to 32 subscribers (e.g., 1:32 split). In the downstream, the OLT transmits downstream traffic through the splitter. Since the splitter passively distributes the optical signal from the OLT to all ONUs/ONTs on the same splitter, each ONU/ONT can physically receive the downstream transmissions destined for other ONUs/ONTs. At the transmission layer, downstream bytes are carried in GPON Transmission Convergence (GTC) frames sent periodically(per-125 μ s), whose payload area carries multiple GEM frames(described next).

METHODOLOGY

- 1. Premise The attacker has an optical capture device (e.g., FPGA with optical TAP, modified ONU that can receive raw GTC frames) connected to splitter and passively records downstream from OLT. • *
- * The attacker does not need to register their ONU/ONT on OLT; so even if they use receive-only optical tap, they can entirely record downstream records without transmitting. • 2. Collect GEMs Then, attacker extracts GTC frames from downstream traffic and parses GEM frames. • 3. Convert GEM PLI to dataset Then, the attacker filters GEM frames for a victim's Port-ID and records PLI values with timestamps. • 4. Dataset format The dataset must include start & end time and duration and PLI and sequence information. • 5. Classification via database matching The attacker matches the captured PLI time series to a pre-collected database of PLI traces (built from known videos) and selects the closest match to infer the victim's video.

EXPERIMENT

we developed a software GPON downstream **simulator** that reproduces the process of receiving GTC frames, extracting GEM frames, and exporting plaintext GEM header fields into datasets. At first we played each of 10 videos three times in our linux based system to produce packet capture datas (PCAPs) where other streaming traffic(e.g., VNC) is restricted to minimize interference. then we simulated that PCAPs packet capture file to records the resulting PLI values and timestamps into CSV traces

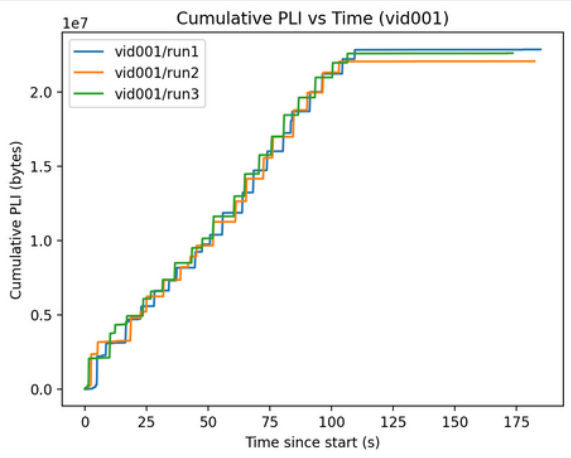


fig. PLI - Time Graph for one of 10 videos showing that greatly identical slopes between three runs, PLI - Time Graph of others is publicly available on our github repository

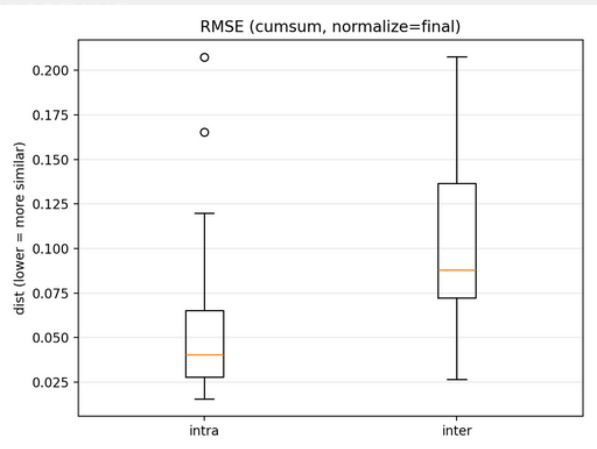


fig.RMSE of Cumulative PLI
left: Same Video
Right: Different Video

CONCLUSION

GPON Network may be vulnerable to streaming inferring attack.
Payload encryption alone does not ensure confidentiality of traffic.
RMSE: 0.040 (Same Video) / 0.088 (Different Video)

FUTURE WORK

Future Work will be conducted in experimental testbed with Real OLT and Optical Splitter. We will use FPGA to collect GEM frames.
when adopted CNN, it is possible to distinguish video Ids using captured PLI values.

DATASET & CODES

Our datasets and codes are publicly available on our github
<https://github.com/hjaka08/GPON>

ACKNOWLEDGEMENTS

I'm a Korean High School Student, please excuse imperfect wording.
Thank you!

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